

Ex.no:9	MINI PROJECT – SOCIAL MEDIA DATA ANALYTICS

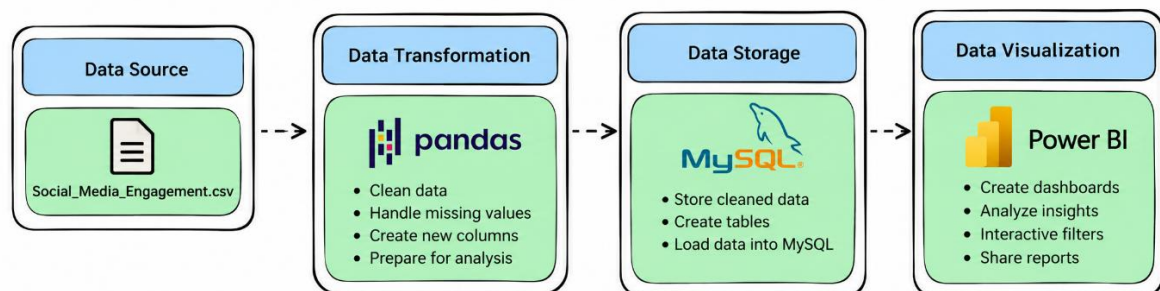
Aim:

To analyze Social Media Data using Tableau by performing data preparation, calculated-field analysis, interactive visualizations, parameters, filters, and dashboards to identify patterns in engagement, reach, content performance, media type, and posting trends.

Algorithm:

1. Load the social media analytics dataset from the CSV file using Pandas.
2. Clean the data by handling missing values, removing duplicate records, and eliminating unnecessary attributes.
3. Transform the data by standardizing columns and deriving useful fields such as total engagement, engagement rate, and performance group.
4. Store the cleaned and transformed dataset in a MySQL database named social_media_analytics.
5. Connect the MySQL database to Tableau for data analysis and reporting.
6. Create calculated fields in Tableau for key metrics such as total posts, total engagement, engagement rate, and performance group.
7. Develop interactive visualizations to analyze engagement trends, content categories, media types, reach, and posting patterns.
8. Apply Tableau parameters, filters, dashboard actions, and interactive features for deeper analysis.
9. Generate business insights from the Tableau dashboard and Story.
10. End the process.

Architecture:



Program:

1. Python – Data Cleaning & Transformation

```
import pandas as pd

from sqlalchemy import create_engine, text

# Load dataset

df = pd.read_csv("instagram_analytics.csv")

# Create a copy for processing

social = df.copy()

# Remove unnecessary columns

# Only remove columns if they exist in your dataset
# social.drop(columns=["unnecessary_column"], inplace=True)

# Handle missing values

social["caption_length"] = social["caption_length"].fillna(0)

social["hashtags_used"] = social["hashtags_used"].fillna(0)

social["traffic_source"] = social["traffic_source"].fillna("Unknown")

social["content_category"] = social["content_category"].fillna("Unknown")

# Remove duplicate records
```

```
social.drop_duplicates(inplace=True)

# Convert timestamp

social["timestamp"] = pd.to_datetime(
    social["timestamp"]
)

# Create derived columns

social["total_engagement"] = (
    social["likes"] +
    social["comments"] +
    social["shares"] +
    social["saves"]
)

social["engagement_rate_calculated"] = (
    social["total_engagement"] /
    social["reach"]
)

# Avoid division problems

social["engagement_rate_calculated"] = (
    social["engagement_rate_calculated"].fillna(0)
)

# Create performance group

social["performance_group"] = pd.cut(
    social["engagement_rate_calculated"],
```

```
bins=[-1, 0.05, 0.10, float("inf")],  
labels=["Low", "Medium", "High"]  
)
```

```
# Standardize column names
```

```
social.columns = (  
    social.columns  
    .str.lower()  
    .str.strip()  
    .str.replace(" ", "_")  
)
```

```
print("Original Shape:", df.shape)
```

```
print("Cleaned Shape:", social.shape)
```

```
social.head()
```

2. MySQL – Database and Data Loading

MySQL Command Prompt

```
net start MySQL80
```

```
mysql -u root -p
```

MySQL

```
CREATE DATABASE IF NOT EXISTS social_media_analytics;
```

Python/Jupyter

```
username = "root"
```

```
password = "YOUR_MYSQL_PASSWORD"
```

```
host = "localhost"
```

```
port = 3306
```

```
# Connect to MySQL database
```

```
db_engine = create_engine(  
    f'mysql+pymysql://{username}:{password}@{host}:{port}/social_media_analytics'  
)
```

```
# Load cleaned data into MySQL
```

```
social.to_sql(  
    "instagram_analytics",  
    con=db_engine,  
    if_exists="replace",  
    index=False,  
    chunksize=5000  
)
```

```
print("Data loaded successfully into MySQL.")
```

Verify

```
pd.read_sql(  
    "SELECT COUNT(*) AS total_rows FROM instagram_analytics",  
    db_engine  
)
```

You should get approximately 30,000 rows.

3. Tableau – Calculated Fields

Total Engagement

[Likes] + [Comments] + [Shares] + [Saves]

Name:

Total Engagement

Calculated Engagement Rate

IF [Reach] > 0 THEN

[Total Engagement] / [Reach]

END

Name:

Calculated Engagement Rate

Performance Group

IF [Calculated Engagement Rate] >= 0.10 THEN

"High"

ELSEIF [Calculated Engagement Rate] >= 0.05 THEN

"Medium"

ELSE

"Low"

END

Name:

Performance Group

Average Reach

No calculated field is required.

In Tableau:

AVG(Reach)

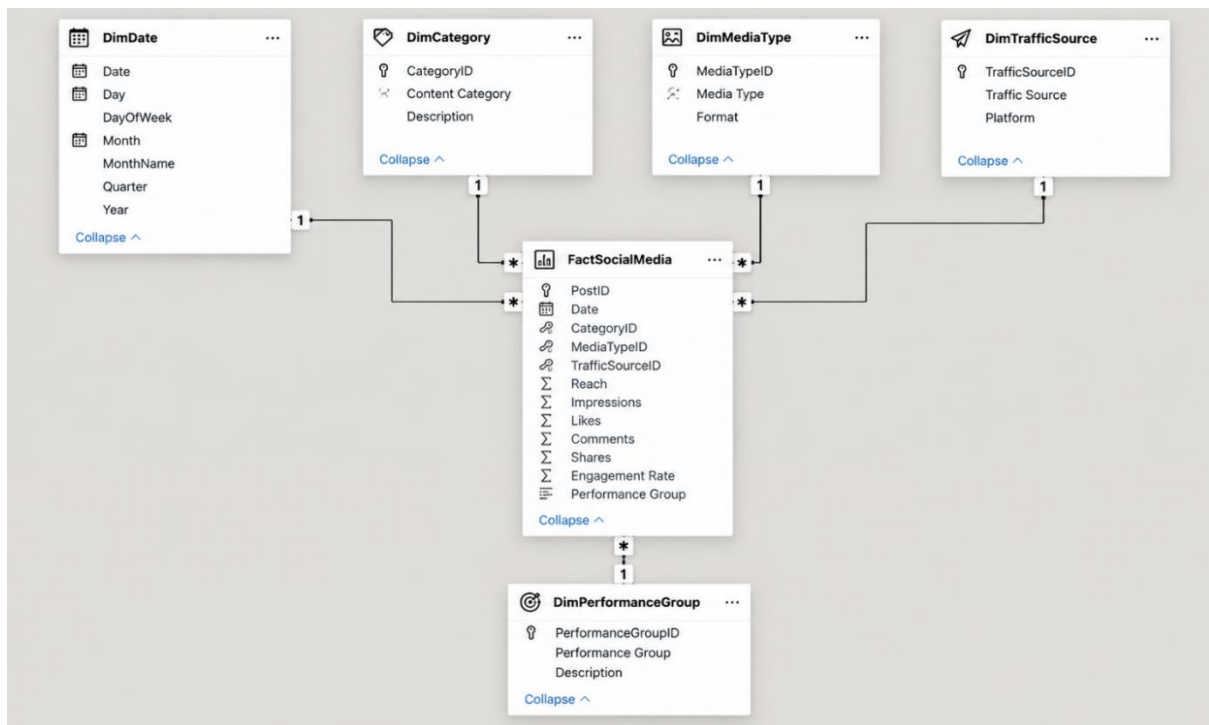
Total Posts

COUNTD(Post ID)

4. Tableau Dashboard

Created visuals in tableau in separate sheets and created a dashboard

Output:



Total Engagement and Average Engagement Rate by Date

1.25M
Goal: 1.25M (+0%)

Total Posts
12,345

Average Engagement Rate
4.72%



Media Type

- ☒ Image
- ☐ Video
- ☐ Carousel
- ☐ Text

Content Category

- ☐ Entertainment
- ☐ Education
- ☒ News
- ☐ Lifestyle
- ☐ Technology
- ☐ Business

Performance Group

- ☐ High
- ☒ Medium
- ☐ Low

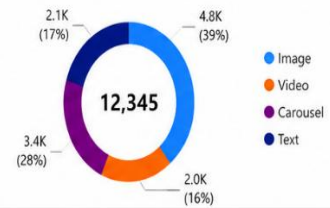
Total Engagement
1.25M

Average Reach
85.3K

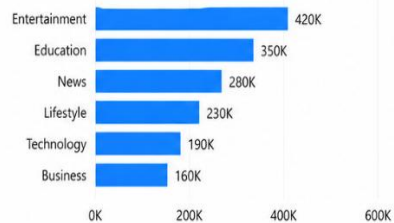
Total Engagement and Average Engagement Rate by Date

1.25M
Goal: 1.25M (+0%)

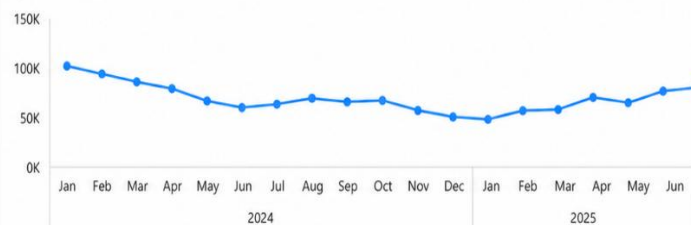
Total Posts by Media Type



Total Engagement by Content Category



Total Engagement by Year and Month



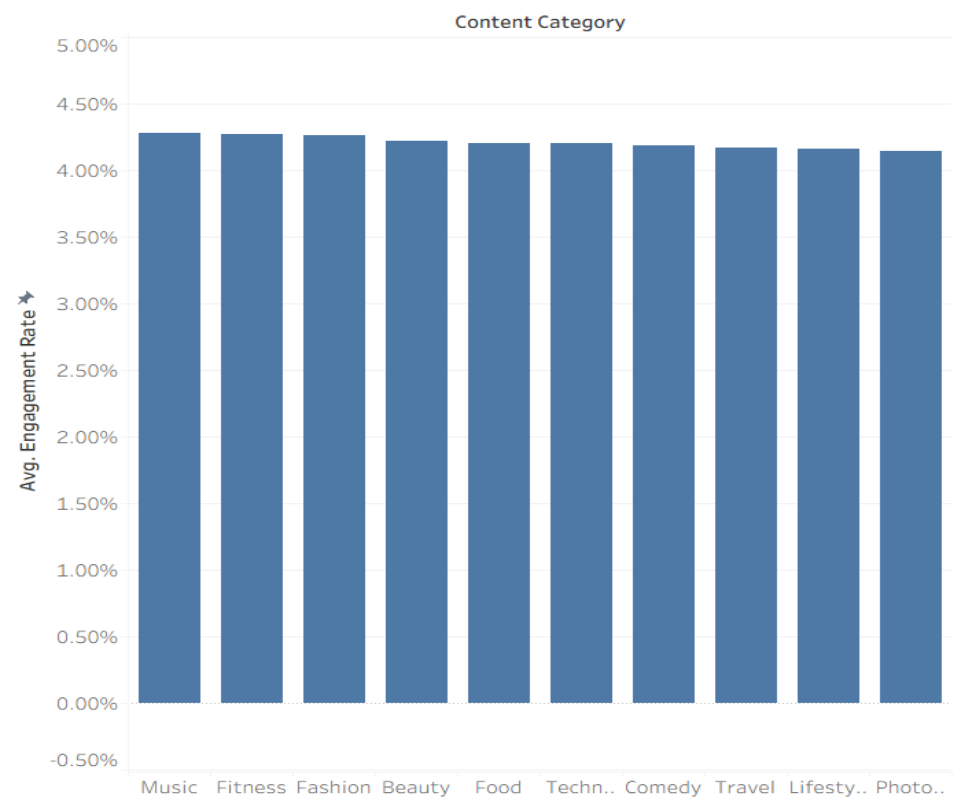
Sheet 3

TOTAL ENGAGEMENT
1,05,93,229

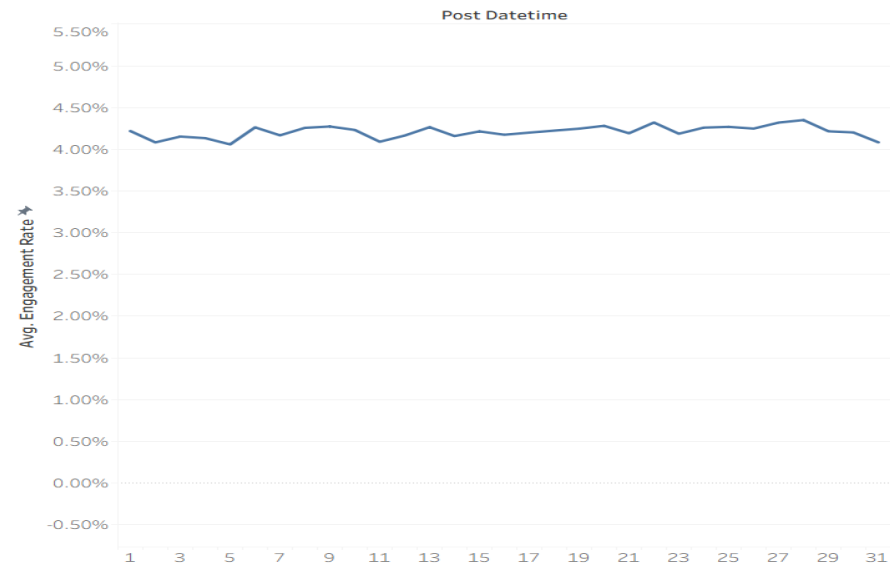
Sheet 4

AVERAGE ENGAGEMENT RATE
4.21%

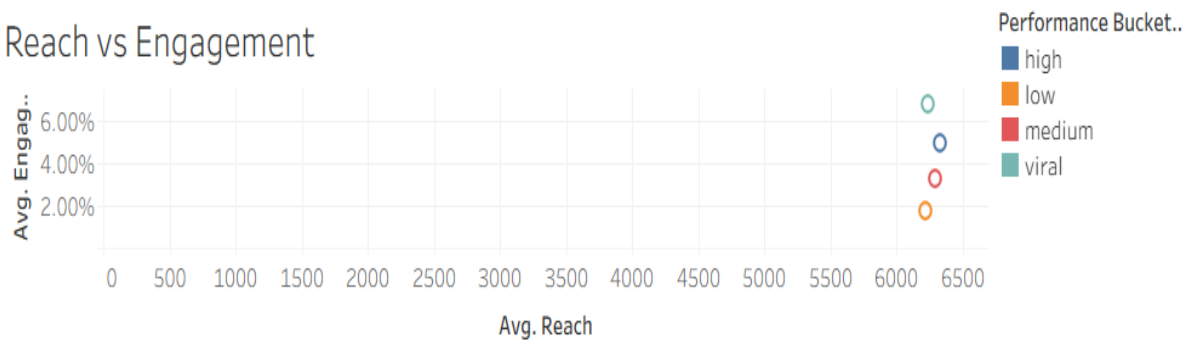
Engagement by Category



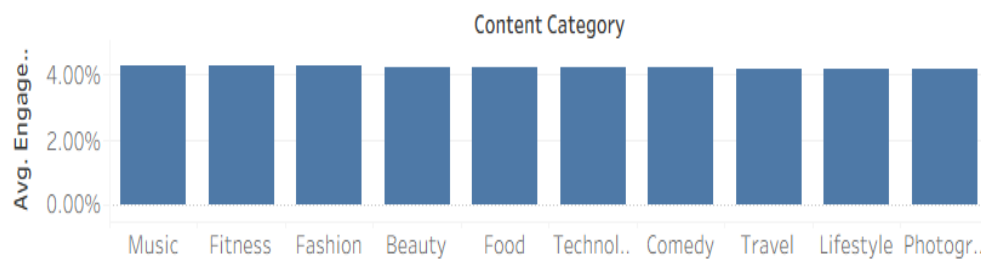
Engagement Trend



Reach vs Engagement



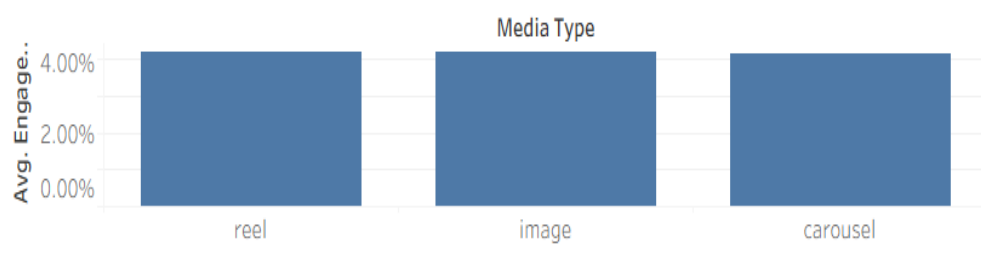
Engagement by Category



Sheet 3

TOTAL ENGAGEMENT
1,05,93,229

Engagement by Media Type



Result: